Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/16/2026
PI Name:	Smith, Amanda L. Ph.D.		
Project Title:	Interface Consistency (Phase 2): Risk Assessment, Standards, and Guidelines		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:		**Solicitation / Funding Source:**	
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Key Personnel Changes/Previous PI:	Dr. Ryan Lange was added to the team in November, 2024, to aid with prototyping.		
COI Name (Institution):	Holden, Kritina Ph.D. (Leidos)		
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Task Description:	Future exploration missions will rely on the design and development of vehicles and complex systems from within NASA and through multiple external commercial partners to meet mission goals. Although consistency-related requirements exist, NASA's current approach to commercial spaceflight development is to allow providers great flexibility and opportunities to innovate. The degree of standardization across Artemis vehicle interfaces is a concern, as a lack of consistency (interface characteristics, interactivity) could impact crew performance and mission success. Results from the Interface Consistency Phase I pilot study suggest inconsistent interfaces are associated with higher workload, increased inefficiency (slower performance time, more actions required), and reduced satisfaction using the system. Determining the level of risk is difficult, as there is no information about the differential impact of specific types of inconsistency related to complex systems. These effects may be exacerbated under the high stress and levels of uncertainty associated with more autonomous future missions. The project team's previous research, Phase I: Characterization and Measurement, addressed many of the shortcomings found in the literature by creating a definition for consistency, formulating a taxonomy of tangible interface characteristics, applying this framework as guidance for developing consistency assessment tools, and conducting an exploratory pilot study to assess performance of these tools. Three prototypes were created for use in the pilot study, collectively exhibiting a range of interface characteristics representative of what crew may use during an Artemis mission and reflecting either high or low consistency with one another. Three different assessment tools were created and used to evaluate differences in consistency between the prototypes. Tools were developed to target different participant groups: the Checklist and Cognitive Walkthrough tools were designed to be used by user interface experts, and the Intersystem Consistency Scale (ICS) was designed to be used by end-user participants with crew-like demographics. This approach provides tool options for use early in the development process (when experts can identify issues for modification), as well as later in the process (when user testing is being performed). Preliminary results from the Phase I pilot study indicate that scores on each of the three assessment tools are able to discriminate between prototype interfaces demonstrating high or low global consistency with a reference prototype and that the tools were judged to be "easy to use". Results also suggest that performance differences, in general, can be observed between consistent and inconsistent interfaces. However, additional work planned for Phase II will: • Refine the assessment tools using feedback and results obtained during Phase I, including: item reduction, item phrasing, adding item descriptions and/or free-form responses, and weighting items or scales, if applicable • Characterize risk by understanding differences in human performance and subjective feedback related to each specific type of inconsistency (i.e., interface characteristics and interaction) • Understand whether this risk is mitigated with practice • Propose a set of standards and guidelines addressing aspects of inconsistent design that appear to have the most impact on performance
Rationale for HRP Directed Research:	Future crewed exploration missions beyond low Earth orbit (LEO), including Artemis, will be made possible by many development efforts from both within NASA and by external partners. The number of different vehicles and systems requiring human interaction will be much greater for future missions. Design solutions across programs and projects are likely to vary, and with NASA's current approach of allowing commercial providers great flexibility in design, the risk of inconsistency is significant. Artemis development efforts are already in progress, which underscores the urgency of this work. This proposal qualifies for directed study due to the insufficient time required for solicitation. Another factor is the highly constrained nature of the work. The proposers are currently involved in commercial programs as subject matter experts, and are very knowledgeable about the diversity of designs currently existing, and the development processes and constraints under which consistency measures and standards would have to be implemented. External researchers would not easily have access to this information; thus, directed research would be more cost effective. This work is leveraging the foundational work completed by the same project team, Interface Consistency Phase I: Characterization and Measurement.
Research Impact/Earth Benefits:	This research aims to better understand human performance impacts that are associated with externally inconsistent interfaces for complex systems. Determining the level of risk is difficult to estimate when reviewing the body of research literature, as there are no "gold standard" tools for assessing the level of consistency between systems and there is no information about the differential impact of specific types of inconsistency related to complex systems. The results could begin to describe this risk and methods that could be used to examine these impacts with complex ground systems and user populations with similar characteristics.
	Aim 1: Refine assessment tools using feedback and results obtained during Phase I, including: item reduction, item phrasing, adding item descriptions and/or free-form responses, and weighting items or scales, if applicable. In addition, provide practitioner documentation. Progress includes: • Selected the Checklist as the expert evaluator assessment tool. This includes refinement of checklist categories, wording, addition of a global consistency scale, and modifying the evaluation process by adding annotation. The tool was tested with human-computer interaction experts at NASA's Ames Research Center and results were incorporated into the final checklist version. A practitioner's guide was also completed, which provides guidance on using the tool and interpreting its results. • The Intersystem Consistency Scale (ICS), the assessment tool developed for crew, was further refined using results obtained during Phase I of the project. This included reduction of redundant items and adding the Intersystem Severity Rubric (ISR) as a complimentary tool to allow raters to describe perceived severity (e.g., performance impacts) associated with the most salient inconsistencies. The ICS and ISR were evaluated internally and by a program interface expert. Results from the Phase II Risk Assessment Study were used to further refine the ICS items, resulting in a final set of 15, and to demonstrate the tools' reliability and validity. As with the checklist, a practitioner guide was developed to aid in using the tools and interpreting their results. Aim 2: Refine tasks and prototype stimuli using feedback from Program human interface and Artemis training Subject Matter Experts. • Interviews were conducted with Training and Program Interface Subject Matter Experts (SMEs) to derive a more

Task Progress:	realistic set of tasks and interface attributes that may be expected for our exploration crews (i.e., across multiple vehicles within a single mission). These interviews were summarized into a report. The insights were used to create a final set of test procedures and to identify interface and interaction attributes that would be included as manipulations across prototypes. • Phase I prototype manipulations were broad, in that each prototype was either “fully consistent” (interface attributes and interactivity) or “fully inconsistent” to the baseline design. The five prototypes for Phase II included more granular design manipulations on individual consistency dimensions. For instance, relative to the “Andromeda” baseline prototype, “Betazed” was designed to be consistent with the interface attributes and interactivity. “Corona” was designed to be inconsistent with the interface attributes, but consistent in interactivity. “Doublestar” was designed to be inconsistent with its interactivity, but consistent with its interface attributes. “Equinox” was designed to be inconsistent with the interface attributes and interactivity of Andromeda. The prototypes were coded in Axure, capturing time on task, click frequency, and accuracy of response to simulated questions from MCC. In addition, training tools were developed which included a PowerPoint slide deck, explaining the color meaning, icons, symbols, interactivity, and a training video, which was a recorded narrated screen capture that demonstrated completion of all procedure tasks on that prototype. The team also created a learning activity, set up as stimulus-response format, to assess participant understanding of the design attributes (e.g. color, icon, symbol meaning) featured in each prototype. This was also coded in Axure, which allowed the capture accuracy and reaction time data for the response to each of the stimuli. Aim 3: Characterize risk by understanding differences in human performance and subjective feedback related to each specific type of inconsistency (i.e., interface characteristics and interaction), low-level interface attributes, and the impact of switching between designs. • A risk assessment study was conducted to provide insight as to when inconsistency matters, using the tasks and prototypes discussed in Aim 2. The study was between-subjects with four conditions; participants were exposed to the baseline design and one additional design (e.g., Andromeda-Betazed, Andromeda-Corona, etc.). 48 participants (12 per condition) were recruited from the Johnson Space Center (JSC) Human Test Subjects Facility and through word of mouth. There were two study sessions: one for training and one for testing the impact on performance related to switching between designs. During the first session, participants received training on the first prototype’s design, followed by the learning activity, and then donned a head-worn eyetracker and completed three procedure trials on that prototype. After the third trial, participants provided ratings on subjective usability and workload associated with completing tasks on that prototype. Then, participants took a quick break and switched to a different monitor in the same room to simulate translating to a different vehicle. They then completed the same activities on a second prototype; the prototype order was counterbalanced. During the second (“Test”) session, participants first completed tasks on the baseline prototype and then switched to another monitor to complete tasks on the other prototype. Performance measures were recorded, including eyetracking measures, clicks, time, and accuracy. Once the last trial was complete, participants rated the consistency of the two prototypes and discussed their ratings during a debrief with the test conductor. Finally, participants provided their demographics and subjective ratings on the usability of the consistency assessment tool within a post-session survey. • Results from the study have been analyzed and summarized into a report. Performance differences were more often found for prototype pairs exhibiting more inconsistency manipulations and fewer performance differences were found within prototype pairs that were most consistent with each other. Responses to stimuli in the learning activity and through the simulated MCC prompts had near-perfect accuracy, but items that were missed were often associated with misinterpretation of item status, likely due to incorrect application of color meaning and/or similarity of features with other icons. Aim 4: Propose a set of standards and guidelines addressing aspects of inconsistent design that appear to have the most impact on performance. • A set of standards and guidelines from industry, government, and agency sources was collected during Phase I of the project. • Results from the Risk Assessment Study guided the team in downselecting from this set and further modifying and/or creating a set of standards and guidelines for potential inclusion into the NASA Standard 3001 Volume 2 and/or other applicable agency documents.
Bibliography Type:	Description: (Last Updated: 04/28/2026)
Abstracts for Journals and Proceedings	Smith A, Holden K, Upasani S, Robertson I, Karasinski J, Lange, R., Mosonyi J, Spanos, Z., Diokno, A. "Interface consistency phase II: Risk assessment, standards & guidelines." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026